

Table 1: A comparison of Kubernetes and Amazon ECS

	Kubernetes	Amazon ECS
Deployment infrastructure	Public clouds or physical hardware	Amazon Web Services EC2 instances
Application definition	Applications are deployed using a combination of pods, deployments and services.	Applications can be deployed as tasks, which are Docker containers running EC2 instances.
Application scalability	Every application tier is regarded as a pod and can be scaled easily when managed by a deployment or replication controller.	An application can be defined using task definitions written in JSON.
High availability	Pods are allocated among worker nodes to provide high availability. This is provided by preventing infrastructure and application failures.	Via the AWS availability zone and the defined placement policies in the service requirements.
Load balancing	Pods are exposed through a service which acts as the load balancer within the cluster.	It can be configured to use elastic load balancing to distribute traffic evenly across all the tasks in the service.
Health monitoring	There are two types of monitoring: Liveness and Readiness.	Health monitoring can be done using CloudWatch.
Block storage	It has two storage APIs: (1) The first provides abstractions for individual storage backends. Example: NFS, Ceph and Flocker. (2) Provides abstraction for storage resource requests.	It is limited to Amazon Web Services.
Networking	The networking model is a flat network, enabling all pods to communicate with one another. The model requires two CIDRs: one from which pods get an IP address, the other for services.	ECS is supported in a VPC, which can include multiple subnets in multiple AZs.
Service discovery	Services can be found using environment variables or DNS.	Services can be found using an ELB and a CNAME.
Performance	Kubernetes scales to 5,000-node clusters. Multiple clusters can be used to scale beyond this limit.	ECS has been scaled-up to over 1000 container nodes without noticeable performance degradation.

- CloudTrail: Logs ECS API calls.
- The advantages of Kubernetes over Amazon ECS are:
- Can be deployed on physical hardware, over public clouds or anywhere on an end user machine.
- Consists of a wide range of storage options like SANs and public clouds.
- Is supported both by Google and Red Hat.
- Has large community support.

The final verdict

Kubernetes is regarded as the better orchestration container tool when compared to Amazon ECS, as it can be deployed anywhere and has more hybrid features with regard to scalability and health checks, as compared to Amazon ECS. 

References

- [1] <https://kubernetes.io/>
- [2] <https://aws.amazon.com/ec2/>
- [3] <https://aws.amazon.com/blogs/aws/cloud-container-management/>

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